MATH 547: HOMEWORK 4
DUE ON: MONDAY, NOVEMBER 4, 9AM.

Problem 1: more on the symmetrization inequality, 30 points:

(1) (De-symmetrization inequality, 15pts) Assume that $X_1, \dots, X_n \in S$ are i.i.d. random variables. Let $\mathcal{F}$ be a class of functions $\mathcal{F} = \{f : S \mapsto \mathbb{R}\}$, and set $Pf := \mathbb{E}f(X)$. Let $\varepsilon_1, \dots, \varepsilon_n$ be i.i.d. Rademacher random variables (i.e., random signs). Following the same steps as we used in the proof of the symmetrization inequality, prove that

$$\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{j=1}^{n}(f(X_i)-Pf)\right| \geq \frac{1}{2}\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{j=1}^{n}\varepsilon_j\,(f(X_j)-Pf)\right|$$

$$\geq \frac{1}{2}\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{j=1}^{n}\varepsilon_j f(X_j)\right| - \frac{1}{2\sqrt{n}}\sup_{f\in\mathcal{F}}Pf.$$

(2) (Symmetrization with Gaussian weights, 15pts) Let $g_1, \dots, g_n$ be i.i.d. $N(0,1)$ random variables. Prove the following version of the symmetrization inequality:

$$\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{j=1}^{n}(f(X_i)-Pf)\right| \leq 2\sqrt{\frac{\pi}{2}}\mathbb{E}\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{j=1}^{n}g_i f(X_i)\right|.$$

[Hint: represent $g_i = \varepsilon_i|g_i|$, where ε_i is independent of $|g_i|$, and use Jensen's inequality]

(1) $\frac{1}{n}\sum_{i=1}^{n} f(X_i) - Pf = \frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathop{\mathbb{E}}_{X'} f(X_i')$

where X_i' is an independent copy of X_i.

$\Rightarrow \frac{1}{n}\sum_{i=1}^{n} f(X_i) - Pf = \frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathop{\mathbb{E}}_{X'} f(X_i')$

$= \mathop{\mathbb{E}}_{X'}\left[\frac{1}{n}\sum_{i=1}^{n} f(X_i) - f(X_i')\right]$

Hence, $\mathop{\mathbb{E}}_{X,\varepsilon}\sup_f \left|\frac{1}{n}\sum_i \varepsilon_i\left(f(X_i) - Pf\right)\right|$

$= \mathop{\mathbb{E}}_{X,\varepsilon}\sup_f \left|\mathop{\mathbb{E}}_{X'}\frac{1}{n}\sum_i \varepsilon_i\left(f(X_i) - f(X_i')\right)\right|$

(Jensen's Ineq.) $\leq \mathop{\mathbb{E}}_{X,\varepsilon}\sup_f \mathop{\mathbb{E}}_{X'}\left|\frac{1}{n}\sum_i \varepsilon_i\left(f(X_i) - f(X_i')\right)\right|$

$$\leq \mathop{\mathbb{E}}_{x,\varepsilon}\mathop{\mathbb{E}}_{x'}\sup_f \left|\frac{1}{n}\sum_i \varepsilon_i\left(f(x_i)-f(x_i')\right)\right|$$

$\because \mathop{\mathbb{E}}_y \sup_z \underbrace{h_z(y)}_{h(z,y)} \geq \sup_z \mathop{\mathbb{E}}_y h_z(y)$

$$= \mathop{\mathbb{E}}_{x,x'}\mathop{\mathbb{E}}_{\varepsilon}\sup_f \left|\frac{1}{n}\sum_i \varepsilon_i\left(f(x_i)-f(x_i')\right)\right|$$

Since $f(x_i)-f(x_i')$ has symmetric distribution,

$\varepsilon_i\left(f(x_i)-f(x_i')\right) \overset{\text{dist}}{=} f(x_i)-f(x_i')$

$$= \mathop{\mathbb{E}}_{x,x'}\frac{1}{2}\sup_f \left|\frac{1}{n}\sum\left(f(x_i)-f(x_i')\right)\right|$$

$$= \mathop{\mathbb{E}}_{x,x'}\sup_f\left\{\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right.\right.$$
$$\left.\left.-\frac{1}{n}\sum_i\left(f(x_i')-Pf\right)\right|\right\}$$

(Δ ineq.) $$\leq \mathop{\mathbb{E}}_{x,x'}\sup_f\left\{\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right|\right.$$
$$\left.+\left|\frac{1}{n}\sum_i\left(f(x_i')-Pf\right)\right|\right\}$$

$$\leq \mathop{\mathbb{E}}_{x,x'}\sup_f\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right|$$

$\because \sup_z g(z)+h(z) \leq \sup_z g(z)+\sup_z h(z)$

$$+\mathop{\mathbb{E}}_{x,x'}\sup_f\left|\frac{1}{n}\sum_i\left(f(x_i')-Pf\right)\right|$$

$$= \mathop{\mathbb{E}}_{x}\sup_f\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right|$$
$$+\mathop{\mathbb{E}}_{x'}\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right|$$

$$= 2\mathop{\mathbb{E}}_{x}\sup_f\left|\frac{1}{n}\sum_i\left(f(x_i)-Pf\right)\right|$$

Hence we get

$$2 \mathop{\mathbb{E}}_{X} \sup_{f} \left| \frac{1}{n} \sum_i \left(f(x_i) - Pf \right) \right|$$

$$\geq \mathop{\mathbb{E}}_{X,\varepsilon} \sup_{f} \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(x_i) - Pf \right) \right|$$

or $\mathop{\mathbb{E}}_{X} \sup \left| \frac{1}{n} \sum_i f(x_i) - Pf \right|$

$$\geq \frac{1}{2} \mathop{\mathbb{E}}_{X,\varepsilon} \sup_{f} \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(x_i) - Pf \right) \right|$$

We will also show this by the general de-symmetrization lemma :—

Let $Z_1, Z_2, \ldots Z_n$ be independent stochastic processes indexed by $t \in T$ such that $\mathbb{E}\, Z_i(t) = 0 \quad \forall\, t \in T$, then

$$\mathbb{E} \sup_{t \in T} \left| \sum_i \varepsilon_i Z_i(t) \right| \leq 2\, \mathbb{E} \sup_{t \in T} \left| \sum_{i=1}^{n} Z_i(t) \right|$$

Proof :- Let $Z_i'(t)$ be an independent copy of $Z_i(t)$.

$$\mathop{\mathbb{E}}_{Z'} Z_i'(t) = 0 \Rightarrow \mathop{\mathbb{E}}_{Z'} \sum_{i=1}^{n} Z_i'(t) = 0$$

$$\sum_{i=1}^{n} Z_i(t) = \sum_{i=1}^{n} Z_i(t) - 0 = \sum_{i=1}^{n} Z_i(t) - \mathop{\mathbb{E}}_{Z'} \sum_{i=1}^{n} Z_i'(t)$$

$$= \mathop{\mathbb{E}}_{Z'} \sum_{i=1}^{n} Z_i(t) - Z_i'(t)$$

Therefore $\sup_{t\in T}\left|\sum_{i=1}^{n} Z_i(t)\right| = \sup_{t\in T}\left|\mathbb{E}_{Z'}\left(Z_i(t) - Z_i'(t)\right)\right|$

$|\cdot|$ is convex, Jensen's ineq. $\leq \sup_{t\in T}\mathbb{E}_{Z'}\left|Z_i(t) - Z_i'(t)\right|$

Sup is convex, Jensen's ineq. $\leq \mathbb{E}_{Z'}\sup_{t\in T}\left|Z_i(t) - Z_i'(t)\right|$

$\mathbb{E}\sup_i Y_i \geq \sup_i \mathbb{E}\, Y_i$

Pf: $\sup_i Y_i \geq Y_j \;\forall j$

$\mathbb{E}\sup_i Y_i \geq \mathbb{E}\, Y_j \;\forall j$

pick the largest

$\mathbb{E}\sup_i Y_i \geq \sup_j \mathbb{E}\, Y_j$

$$\Rightarrow \quad \mathbb{E}_{Z}\sup_t\left|\sum Z_i(t)\right| \leq \mathbb{E}_{Z}\mathbb{E}_{Z'}\sup_t\left|\sum_i Z_i(t) - Z_i'(t)\right|$$

Since $Z_i(t) - Z_i'(t) \overset{\text{dist.}}{=} Z_i'(t) - Z_i(t) = -\left(Z_i(t) - Z_i'(t)\right)$,

multiplying by a Radamacher random variable ε_i does not change the distribution, i.e. $\varepsilon_i\left(Z_i(t) - Z_i'(t)\right) \overset{\text{dist}}{=} Z_i(t) - Z_i'(t)$

$$\therefore \quad \mathbb{E}_{Z,Z'}\sup_t\left|\sum_i\left(Z_i(t) - Z_i'(t)\right)\right| = \mathbb{E}_{Z,Z',\varepsilon}\left|\sum_i \varepsilon_i\left(Z_i(t) - Z_i'(t)\right)\right|$$

Proof: $\mathbb{E}_{Z,Z'}\mathbb{E}_{\varepsilon}\left|\sum_i \varepsilon_i\left(Z_i(t) - Z_i'(t)\right)\right| = \frac{1}{2}\mathbb{E}_{Z,Z'}\left|\sum_i\left(Z_i(t) - Z_i'(t)\right)\right|$

$$+ \frac{1}{2}\mathbb{E}_{Z,Z'}\left|\sum_i\left(Z_i'(t) - Z_i(t)\right)\right|$$

$$= \frac{1}{2}\mathbb{E}_{Z,Z'}\left|\sum_i\left(Z_i(t) - Z_i'(t)\right)\right|$$

$$+ \frac{1}{2}\mathbb{E}_{Z,Z'}\left|-\sum_i\left(Z_i(t) - Z_i'(t)\right)\right|$$

$$= 2\cdot\frac{1}{2}\mathbb{E}_{Z,Z'}\left|\sum_i\left(Z_i(t) - Z_i'(t)\right)\right|$$

$$= \mathbb{E}_{Z,Z'}\left|\sum_i\left(Z_i(t) - Z_i(t)\right)\right|$$

Hence $\mathbb{E}_{Z} \sup_t \left| \sum_i Z_i \right| \le \mathbb{E}_{Z} \mathbb{E}_{Z'} \sup_t \left| \sum_i (Z_i(t) - Z_i'(t)) \right|$

$$= \mathbb{E}_{Z,Z',\varepsilon} \sup_t \left| \sum_i \varepsilon_i (Z_i(t) - Z_i'(t)) \right|$$

$$\Rightarrow \mathbb{E}_{Z} \sup_t \left| \sum_i Z_i \right| \le \mathbb{E}_{Z,Z',\varepsilon} \sup_t \left| \sum_i \varepsilon_i (Z_i(t) - Z_i'(t)) \right|$$

$$\le \mathbb{E}_{Z,Z',\varepsilon} \sup_t \left\{ \left| \sum_i \varepsilon_i Z_i(t) \right| + \left| \sum_i \varepsilon_i Z_i'(t) \right| \right\}$$

Δ ineq.
(De-Symmetrization step)

$$\le \mathbb{E}_{Z,Z',\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i(t) \right| + \mathbb{E}_{Z,Z',\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i'(t) \right|$$

$$= \mathbb{E}_{Z,\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i(t) \right| + \mathbb{E}_{Z',\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i'(t) \right|$$

$$= 2 \, \mathbb{E}_{Z,\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i(t) \right|$$

$$\therefore \quad \mathbb{E}_{Z} \sup_t \left| \sum_i Z_i(t) \right| \le 2 \, \mathbb{E}_{Z,\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i(t) \right|$$

$$\text{or} \quad \mathbb{E}_{Z,\varepsilon} \sup_t \left| \sum_i \varepsilon_i Z_i(t) \right| \ge \frac{1}{2} \mathbb{E}_{Z} \sup_t \left| \sum_i Z_i(t) \right|$$

To prove the statement in question, we observe that $Z = X$ and $Z(f) = X(f) = \frac{1}{n}\left(f(x_i) - Pf\right)$

and this does satisfy the premise of the de-symmetrization lemma :-

$$\mathbb{E}\, Z(f) = \mathbb{E}\, f(x_i) - Pf = Pf - Pf = 0$$

Then the de-symmetrization lemma says

$$\mathop{\mathbb{E}}_{Z,\varepsilon} \sup_f \left| \sum_i \varepsilon_i Z_i(f) \right| \geq \frac{1}{2} \mathop{\mathbb{E}}_{Z} \sup_f \left| \sum_i Z_i(f) \right|$$

which gives :-

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(x_i) - Pf \right) \right|$$

$$\geq \frac{1}{2} \mathop{\mathbb{E}}_{X} \sup_f \left| \frac{1}{n} \sum_i \left(f(x_i) - Pf \right) \right|$$

Now, we want to show the second inequality asked in the question :-

$$\frac{1}{2} \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i \left(f(x_i) - Pf \right) \right|$$

$$\geq \frac{1}{2} \left\{ \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(x_i) \right| - \frac{1}{\sqrt{n}} \sup Pf \right\}$$

First, we will show that

$$\sup_f |a_f - b_f| \geq \sup_f |a_f| - \sup_f |b_f|$$

Proof: $\forall f,\ |a_f| = |a_f - b_f + b_f| \leq |a_f - b_f| + |b_f|$ (Δ ineq.)

$$\Rightarrow |a_f| - |b_f| \leq |a_f - b_f| \quad \forall f$$

$$\Rightarrow |a_f| - |b_f| \leq \sup_f |a_f - b_f| .$$

$$\Rightarrow |a_f| - \sup_f |b_f| \leq \sup_f |a_f - b_f|, \ \forall f$$

Since the above holds for all f, we can also take the particular f that maximizes $|a_f|$.

Hence $\sup_f |a_f| - \sup_f |b_f| \leq \sup |a_f - b_f|$

Now $\sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(X_i) - Pf \right) \right|$

$$\geq \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \underbrace{\sup_f \left| \frac{1}{n} \sum_i \varepsilon_i Pf \right|}_{\sup_f \left| Pf \cdot \frac{1}{n} \sum_i \varepsilon_i \right|}$$

$$= \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \underbrace{\sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \right| \cdot |Pf|}_{\left| \frac{1}{n} \sum_i \varepsilon_i \right| \cdot \sup_f |Pf|}$$

Taking expectation both sides

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(x_i) - Pf \right) \right|$$

$$\geq \mathop{\mathbb{E}}_{X,\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right| \sup_f |Pf|$$

$$= \underbrace{\sup_f |Pf|}_{\text{Constant, not random w.r.t } X \text{ or } \varepsilon} \mathop{\mathbb{E}}_{X,\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right|$$

Now, $\mathop{\mathbb{E}}_{\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right| = \mathop{\mathbb{E}}_{\varepsilon} \sqrt{\left| \frac{1}{n} \sum_i \varepsilon_i \right|^2}$

($\sqrt{\cdot}$ is concave, Jensen's ineq.) $\leq \sqrt{\mathop{\mathbb{E}}_{\varepsilon} \left| \frac{1}{n} \sum \varepsilon_i \right|^2}$

$$\leq \sqrt{\frac{1}{n^2} \mathop{\mathbb{E}}_{\varepsilon} \left| \sum_i \varepsilon_i \right|^2}$$

$$= \sqrt{\frac{1}{n^2} \mathop{\mathbb{E}}_{\varepsilon} \left(\sum_i \varepsilon_i^2 + \sum_{i \neq j} \varepsilon_i \varepsilon_j \right)}$$

$$= \sqrt{\frac{1}{n^2} \left(\mathop{\mathbb{E}}_{\varepsilon} \sum_i \varepsilon_i^2 + \mathop{\mathbb{E}}_{\varepsilon} \sum_{i \neq j} \varepsilon_i \varepsilon_j \right)}$$

Independence of ε_i & ε_j $= \sqrt{\frac{1}{n^2} \left(\sum_i \mathop{\mathbb{E}}_{\varepsilon_i} \varepsilon_i^2 + \sum \underbrace{\mathop{\mathbb{E}}_{\varepsilon_i} \varepsilon_i}_{0} \underbrace{\mathop{\mathbb{E}}_{\varepsilon_j} \varepsilon_j}_{0} \right)}$

$$= \sqrt{\frac{1}{n^2} (n + 0)} = \frac{1}{\sqrt{n}}$$

$$\therefore \quad \mathop{\mathbb{E}}_{\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right| \leq \frac{1}{\sqrt{n}}$$

Hence,

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(X_i) - Pf \right) \right|$$

$$\geq \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \sup_f |Pf| \cdot \mathop{\mathbb{E}}_{X,\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right|$$

doesn't depend on X

$$= \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \sup_f |Pf| \cdot \mathop{\mathbb{E}}_{\varepsilon} \left| \frac{1}{n} \sum_i \varepsilon_i \right| \cdot \mathop{\mathbb{E}}_{X}(1)$$

independence of ε & X

$$\geq \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \frac{1}{\sqrt{n}} \sup_f |Pf|$$

* Ikle have almost proven the statement except that we have $|Pf|$ instead of Pf.

* I believe this requires additional assumption than given in the question. If $f : S \to \mathbb{R}$ & $f \geq 0$, then $\sup_f |Pf| = \sup_f Pf$

Then we get

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i \left(f(X_i) - Pf \right) \right|$$

ε &

$$\geq \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| - \frac{1}{\sqrt{n}} \sup_f |Pf|$$

Multiplying $\frac{1}{2}$ on both sides gives us the result asked in question.

(2) (Symmetrization with Gaussian weights, 15pts) Let $g_1, \ldots, g_n$ be i.i.d. $N(0,1)$ random variables. Prove the following version of the symmetrization inequality:

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{j=1}^n (f(X_i) - Pf) \right| \leq 2\sqrt{\frac{\pi}{2}} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{j=1}^n g_i f(X_i) \right|.$$

[Hint: represent $g_i = \varepsilon_i |g_i|$, where ε_i is independent of $|g_i|$, and use Jensen's inequality]

We will start with the symmetrization using Rademacher r.v.

$$\mathop{\mathbb{E}}_{X} \sup_f \left| \frac{1}{n} \sum_{i=1}^n \left(f(X_i) - Pf \right) \right| = \mathop{\mathbb{E}}_{X} \sup_f \left| \frac{1}{n} \sum_{i=1}^n \left(f(X_i) - \mathop{\mathbb{E}}_{X'} f(X_i') \right) \right|$$

$$= \mathop{\mathbb{E}}_{X} \sup_f \left| \mathop{\mathbb{E}}_{X'} \frac{1}{n} \sum_{i=1}^n \left(f(X_i) - f(X_i') \right) \right| \quad \text{Symmetrization}$$

Jensen's Ineq. $$\leq \mathop{\mathbb{E}}_{X} \sup_f \mathop{\mathbb{E}}_{X'} \left| \frac{1}{n} \sum_{i=1}^n \left(f(X_i) - f(X_i') \right) \right|$$

Jensen's Ineq. $$\leq \mathop{\mathbb{E}}_{X} \mathop{\mathbb{E}}_{X'} \sup_f \left| \frac{1}{n} \sum_{i=1}^n \left(f(X_i) - f(X_i') \right) \right|$$

$$= \mathop{\mathbb{E}}_{X,X'} \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \left(f(X_i) - f(X_i') \right) \right|$$

Δ ineq $$\leq \mathop{\mathbb{E}}_{X,X',\varepsilon} \sup_f \left\{ \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i) \right| + \left| \frac{1}{n} \sum_i \varepsilon_i f(X_i') \right| \right\}$$

$$= \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum \varepsilon_i f(X_i) \right| + \mathop{\mathbb{E}}_{X',\varepsilon} \sup_f \left| \sum_i \varepsilon_i f(X_i) \right|$$

$$= 2 \underset{X,\varepsilon}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum \varepsilon_i f(x) \right|$$

$$\therefore \underset{X}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - Pf) \right| \leq 2 \underset{X,\varepsilon}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(x_i) \right|$$

→ (1)

Now, we will show that

$$\underset{\varepsilon}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(x_i) \right| \leq \sqrt{\frac{\pi}{2}} \underset{g}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right|$$

$$\underset{g}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right| = \underset{|g|, \varepsilon_i}{\mathbb{E}} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i |g_i| f(x_i) \right|$$

using the hint from the question.

Define $h(u) = \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \sum_i \varepsilon_i u_i f(x_i) \right|$

'h' satisfies the following two properties :-

i) Positive homogeneity: $h(ku) = k\,h(u) \quad \forall\, k > 0$

Proof :- $h(ku) = \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \sum_i \varepsilon_i k u_i f(x_i) \right|$

$$= c \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \sum_i \varepsilon_i u_i f(x_i) \right|$$

$$= c\,h(u)$$

(ii) Convexity : $h(\alpha a + \beta b) \leq \alpha h(a) + \beta h(b)$

Proof: $f(\alpha u_1 + \beta u_2) = \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \sum_i \varepsilon_i (\alpha a_i + \beta b_i) f(x_i) \right|$

$$= \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \sum_i \{ \alpha \varepsilon_i a_i f(x_i) + \beta \varepsilon_i b_i f(x_i) \} \right|$$

$$= \underset{\varepsilon}{\mathbb{E}} \sup_f \left| \alpha \sum_i \varepsilon_i a_i f(x_i) + \beta \sum_i \varepsilon_i b_i f(x_i) \right|$$

$$(\Delta \text{ ineq.}) \quad \leq \alpha \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \sum_i \varepsilon_i a_i f(x_i) \right| + \beta \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \sum_i \varepsilon_i b_i f(x_i) \right|$$

$$= \alpha h(a) + \beta h(b).$$

$$\therefore \mathop{\mathbb{E}}_{g} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right| = \mathop{\mathbb{E}}_{|g|} \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i |g_i| \varepsilon_i f(x_i) \right|$$

$$= \mathop{\mathbb{E}}_{|g|} h(|g|)$$

Observe that $h(\vec{1}) = \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(x_i) \right|$

Hence we want to show that $h(\vec{1}) \leq \sqrt{\frac{2}{\pi}} \mathop{\mathbb{E}}_{|g|} h(|\vec{g}|)$

$$h(|\vec{g}|) = \mathop{\mathbb{E}}_{|g|} \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i |g_i| f(x_i) \right|$$

We note that $\mathop{\mathbb{E}}_{|g_i|} |g_i| = \sqrt{\frac{2}{\pi}} =: m$

Then $\frac{1}{m} \mathop{\mathbb{E}}_{|g_i|} |g_i| = \mathop{\mathbb{E}}_{|g_i|} \left| \frac{g_i}{m} \right|$

$$\therefore \mathop{\mathbb{E}}_{|g|} \frac{|g|}{m} = [1, 1, \ldots 1]^T = \vec{1}$$

where $|\vec{v}| = [|v_1| \ldots |v_n|]^T$ (component wise)

Now $\mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(x_i) \right| = h(\vec{1}) = h\left(\mathop{\mathbb{E}}_{|g|} \frac{|g|}{m} \right)$

($\because h$ is convex, Jensens Ineq.) $\leq \mathop{\mathbb{E}}_{|g|} h\left(\frac{|g|}{m} \right)$

($\because$ positive homogeneity of h) $= \mathop{\mathbb{E}}_{|g|} \frac{1}{m} h(|g|)$

$$= \frac{1}{m} \mathop{\mathbb{E}}_{|g|} h(|g|)$$

$$= \frac{1}{m} \mathop{\mathbb{E}}_{|g|} \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i |g_i| f(x_i) \right|$$

$$= \sqrt{\frac{\pi}{2}} \mathop{\mathbb{E}}_{g} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right|$$

$$\Rightarrow \mathop{\mathbb{E}}_{\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(x_i) \right| \leq \sqrt{\frac{\pi}{2}} \mathop{\mathbb{E}}_{g} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right| \quad \text{→ (2)}$$

Taking expectation on both sides with respect to X

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(x_i) \right| \leq \sqrt{\frac{\pi}{2}} \mathop{\mathbb{E}}_{X,g} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right| \quad \text{→ (2)}$$

From (1) & (2) :-

$$\mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i (f(x_i) - Pf) \right| \leq 2 \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f \left| \frac{1}{n} \sum_i \varepsilon_i f(x_i) \right|$$

$$\leq 2 \sqrt{\frac{\pi}{2}} \mathop{\mathbb{E}}_{X,g} \sup_f \left| \frac{1}{n} \sum_i g_i f(x_i) \right|$$

∎

Problem 3: Rademacher complexity of a ball, 15 points:

Let $\mathcal{F}$ be a d-dimensional space of functions, and $B(r) = \{f \in \mathcal{F} : \|f\|_{L_2(P)} = \mathbb{E}f^2(X) \leq r\}$. Show that

$$\mathbb{E} \sup_{f \in B(r)} |R_n(f)| \leq r\sqrt{\frac{d}{n}},$$

where $R_n(f) = \frac{1}{n}\sum_{j=1}^{n} \varepsilon_j f(X_j)$ is the Rademacher process.

[Hint: use linearity of the Rademacher process together with the fact that any $f \in \mathcal{F}$ can be written as $f = \sum_{j=1}^{d} \alpha_j(f)\phi_j$ where $\phi_1, \ldots, \phi_d$ are the basis functions; you do not need to use generic chaining or entropy integral bounds in this problem!]

Since $\mathcal{F}$ is d-dimensional, we have some basis $\{\phi_1, \ldots \phi_d\}$ for $\mathcal{F}$ such $\forall f \in \mathcal{F}$, $\exists\ \alpha_1, \alpha_2, \ldots \alpha_d \in \mathbb{R}$ such that

$$f = \sum_{i=1}^{d} \alpha_i(f)\,\phi_i$$

Also because the d-dimensional subspace of $L_2(P)$ is also an inner product space, we can find ϕ_i such that they are orthonormal, i.e. $\langle \phi_i, \phi_j \rangle = \delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$

In this orthonormal basis $\alpha_i(f) = \langle f, \phi_i \rangle$

Note: $\langle h, g \rangle = \mathop{\mathbb{E}}_{X}[h(X)\,g(X)]$

$$\|f\|_{L_2(P)} \langle f, f \rangle = \Big\langle \sum_{i=1}^{d} \alpha_i \phi_i, \sum_{i=1}^{d} \alpha_i \phi_i \Big\rangle = \sum_{ij} \alpha_i \alpha_j \langle \phi_i, \phi_j \rangle$$

$$= \sum_{ij} \alpha_i \alpha_j \delta_{ij} = \sum_{ij} \alpha_i^2 = |\vec{\alpha}_f|_2^2$$

$$B(r) = \{ f \in \mathcal{F} : \|f\|_{L_2(P)} \leq r \} = \{ f \in \mathcal{F} : \|\alpha_f\|_2 \leq r \}$$

$$= \{ \alpha \in \mathbb{R} : \|\alpha_f\|_2 \leq r \}$$

using the orthonormal basis $\phi_1, \ldots \phi_d$

$$R_n(f) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i f(x_i) = \frac{1}{n}\sum_{i} \varepsilon_i \sum_{j=1}^{d} \alpha_j \phi_j(x_i)$$

$$= \sum_{j=1}^{d} \alpha_j \underbrace{\frac{1}{n}\sum_{i=1}^{n} \varepsilon_i \phi_j(x_i)}_{z_j}$$

$$= \sum_{j=1}^{d} \alpha_j z_j = \langle \alpha_f, z \rangle$$

$$\sup_f |R_n(f)| = \sup_f \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(x_i) \right| = \sup_{\|\alpha_f\|_2 \le r} |\langle \alpha_f, Z \rangle|$$

$$= r \|Z\|_2$$

$$\Rightarrow \mathop{\mathbb{E}}_{X,\varepsilon} \sup_f |R_n(f)| = \mathop{\mathbb{E}}_{X,\varepsilon} r\|Z\|_2 = r \mathop{\mathbb{E}}_{X,\varepsilon} \|Z\|_2$$

$$= r \mathop{\mathbb{E}}_{X,\varepsilon} \sqrt{\|Z\|_2^2} \le r \sqrt{\mathop{\mathbb{E}}_{X,\varepsilon} \|Z\|^2}$$

Jensen's Ineq.
$\sqrt{\cdot}$ is concave

$$\Rightarrow \mathop{\mathbb{E}}_{X,\varepsilon} |R_n(f)| \le r \sqrt{\mathop{\mathbb{E}}_{X,\varepsilon} \sum_{j=1}^d Z_j^2}$$

$$= r \sqrt{\sum_{j=1}^d \mathop{\mathbb{E}}_{X,\varepsilon} Z_j^2}$$

$$= r \sqrt{\sum_{j=1}^d \frac{1}{n}} = r \sqrt{\frac{d}{n}}$$

$$\mathop{\mathbb{E}}_{X,\varepsilon} Z_j^2 = \mathop{\mathbb{E}}_{X,\varepsilon} \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i \phi_j(x_i) \right)^2$$

$$= \frac{1}{n^2} \mathop{\mathbb{E}}_{X,\varepsilon} \left[\left(\sum_{i=1}^n \underbrace{\varepsilon_i^2}_{1} \phi_j^2(x_i) \right) \right] + \frac{1}{n^2} \mathop{\mathbb{E}}_{X,\varepsilon} \left[\sum_{i=1}^n \sum_{k=1}^n \varepsilon_i \varepsilon_k \phi(x_i) \phi(x_k) \right]$$

$$= \frac{1}{n^2} \left(\mathop{\mathbb{E}}_{X,\varepsilon} \sum_{i=1}^n \phi_j^2(x_i) + \mathop{\mathbb{E}}_{X} \sum_{i=1}^n \sum_{k=1}^n \underbrace{\left(\mathop{\mathbb{E}}_{\varepsilon_i} \varepsilon_i \right)}_{0} \underbrace{\left(\mathop{\mathbb{E}}_{\varepsilon_k} \varepsilon_k \right)}_{0} \phi(x_i) \phi(x_k) \right)$$

$$= \frac{1}{n^2} \left(\mathop{\mathbb{E}}_{X,\varepsilon} \sum_{i=1}^n \phi_j^2(x_i) \right) = \frac{1}{n^2} \mathop{\mathbb{E}}_{X} \sum_{i=1}^n \phi_j^2(x_i) = \frac{1}{n^2} \sum_{i=1}^n \mathop{\mathbb{E}}_{X} \phi_j^2(x_i)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \mathop{\mathbb{E}}_{X_i} \phi_j^2(x_i)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \underbrace{\mathop{\mathbb{E}}_{X_i} [\phi_j(x_i) \phi_j(x_j)]}$$

$$= \frac{1}{n^2} \sum_{i=1}^n \langle \phi_j, \phi_j \rangle = \frac{1}{n} \sum_{i=1}^n \delta_{ii}$$

$$= \frac{1}{n^2} \sum_{i=1}^n 1 = \frac{1}{n^2} \cdot n = \frac{1}{n}$$

Problem 2: facts about VC dimension, 30 points:

(a) (10pts) Let $\mathcal{C}_1, \ldots, \mathcal{C}_k$ be k classes of sets such that $3 \leq d = \max_{j=1,\ldots,k} \mathrm{VC}(\mathcal{C}_j) < \infty$. Prove that

$$\mathrm{VC}\left(\bigcup_{j=1}^{k} \mathcal{C}_j\right) \leq 4d\log(2d) + 2\log(k)$$

If you prove the bound with different constants in place of 2 and 4, you will also receive full credit.

(b) (10pts) Prove directly, without using Dudley's theorem, that the VC dimension of the set of all halfspaces in $\mathbb{R}^d$ is equal to $d+1$.
[Hint: use Radon's theorem which states that any collection of $d+2$ points in $\mathbb{R}^d$ can be partitioned into 2 (disjoint) subsets such that the convex hulls of these subsets intersect]

(c) (10pts) Use part (b) to show that the VC dimension of balls in $\mathbb{R}^d$ is also $d+1$. *[Hint: suppose that the set of points is shattered by the spheres. Then for any partition of this set into 2 disjoint subsets, there exist two spheres such that each contains only the points only from one of these disjoint subsets. Show that there must also be a hyperplane such that the corresponding half-spaces have the same property. This will give you an upper bound for the VC dimension]*

(d) (*bonus + 20pts)

(1) (10 bonus pts) Find an upper and lower bounds for VC dimension of ellipsoids (an ellipsoid is a set of points $x \in \mathbb{R}^d$ that satisfy $(x-x_0)^T A(x-x_0) \leq 1$, where $x_0 \in \mathbb{R}^d$ is fixed and $A \in \mathbb{R}^{d\times d}$ is a positive definite matrix.

(2) (10 bonus pts) Find an upper and lower bounds for VC dimension spherical cones in $\mathbb{R}^d$. A spherical cone is a set

$$C_u(t;b) = \left\{x \in \mathbb{R}^d : \ \langle x-b, u\rangle \geq t\|x-b\|_2\}, \ \|u\|_2 = 1, \ b \in \mathbb{R}^d, \ t \in (0,1]\right\}.$$

a) Let $m_H(n)$ be the growth function of class H

The growth function $m_H(n)$ counts the maximum binary vector labelings that are realizable by $h \in H$ over any n points.

$$m_H(n) = \max_{x_1, x_2 \ldots x_n \in X} \#\left\{[h(x_1), \ldots h(x_n)]^T \;\middle|\; h \in H\right\}$$

Clearly $m_H(n) \leq 2^n$

If $VC(H) \leq d$, then $\forall n \geq d$

$$m_H(n) \leq \sum_{i=0}^{d} \binom{n}{i} \leq \left(\frac{en}{d}\right)^d$$ [Sauer's Lemma]

Now, we are given that $C = \bigcup_{i=1}^{k} C_i$ is a union of k hypothesis classes.

$$VC(C_j) = d_j \quad \& \quad \max_{j=1\ldots k} d_j = d \geq 3$$

By Sauer's Lemma,

$$m_{C_j}(n) \leq \left(\frac{en}{d_i}\right)^j$$

Also let $Y_j(x_{1:n}) = \{[h(x_1) \cdots h(x_n)] \mid h \in C_j\}$

$$\therefore m_{C_j}(n) = \max_{x_{1:n}} \# Y_j(x_{1:n})$$

Let $Y(x_{1:n}) = \{ h(x_1) \cdots h(x_n) \mid h \in C = \bigcup_{j=1}^{n} C_j \}$

For a given set of n points $x_{1:n} := x_1 \ldots x_n$

$$Y(x_{1:n}) = \{ h(x_1) \cdots h(x_n) \mid h \in C = \bigcup_{j=1}^{n} C_j \}$$

$$\subseteq \bigcup_{j=1}^{n} \{ h(x_1) \cdots h(x_n) \mid h \in C_j \}$$

$$= \bigcup_{j=1}^{n} Y_j(x_{1:n})$$

$$\Rightarrow \# Y(x_{1:n}) \leq \sum_{j=1}^{n} \# Y_j(x_{1:n})$$

union bound

$$\Rightarrow \underbrace{\max_{x_{1:n}} \#Y(x_{1:n})}_{m_C(n)} \leq \max_{x_{1:n}} \sum_{j=1}^{k} \#Y_j(x_{1:n})$$

$$\leq \sum_{j=1}^{k} \max_{x_{1:n}} \#Y_j(x_{1:n})$$

$$= \sum_{j=1}^{k} m_{C_j}(n)$$

$$\Rightarrow m_C(n) \leq \sum_{j=1}^{k} \underbrace{m_{C_j}(n)}_{\leq \sum_{p=1}^{d_j} \binom{n}{p}}$$

$$\leq \sum_{j=1}^{k} \sum_{p=1}^{d_j} \binom{n}{p} \quad \text{By Sauer's Lemma}$$

$$\leq \sum_{j=1}^{k} \sum_{p=1}^{d} \binom{n}{p} \quad \text{Since } d \geq d_j \ \forall j$$

$$\leq \sum_{j=1}^{k} \left(\frac{en}{d}\right)^d$$

$$= k\left(\frac{en}{d}\right)^d$$

If $VC(C) \geq n$, then $m_C(n) = 2^n$

Equating both sides

$$2^n = k\left(\frac{en}{d}\right)^d$$

$$\Rightarrow n = \log_2 k + d(\log_2 e + \log_2 n - \log_2 d)$$

$$\Rightarrow n = \log_2 k + d + d\log_2 n - d\log_2 d$$

$$\leq \log_2 k + d + d\log_2 n$$

CASE 1: $n \leq 2d \log_2 2d$

Since $d \geq 3$, $2d < d^2$

$$\therefore n \leq 2d \log_2 d^2 = 4d \log_2 d \leq 4d \log_2 2d$$

$$\leq 4d \log_2 2d + 2 \log_2 k$$

CASE 2: $n > 2d \log_2 2d$

This implies $\log_2 n \leq \frac{n}{2d}$ (plugging & checking, it is true)

Plugging into $n \leq \log_2 k + d + d \log_2 n$

$$n \leq \log_2 k + d + d\frac{n}{2d}$$

$$= \log_2 k + d + \frac{n}{2}$$

$$\Rightarrow \frac{n}{2} \leq \log_2 k + d$$

$$\Rightarrow n \leq 2\log_2 k + 2d$$

Since $d \geq 3$, $d \log_2 d \geq d$

$$\Rightarrow n \leq 2\log_2 k + 2d\log_2 d$$

$$\leq 2\log_2 k + 4d\log_2 d$$

$$\leq 2\log_2 k + 4d\log_2 2d$$

$$\Rightarrow n \leq 4d\log_2 2d + 2\log k$$ ∎

b) Radon's Theorem: Any $d+2$ points in $\mathbb{R}^d$ can be split into two disjoint subsets such that the convex hull of these subsets intersect.

Let $x_1 \ldots x_{d+2}$ be $d+2$ points in $\mathbb{R}^d$.

$\exists$ index sets A & B such that $A \cap B = \phi$ & $A \cup B = \{1, 2, \ldots d+2\}$

$$\left\{ \sum_{i \in A} \alpha_i x_i \;\middle|\; \alpha_i \geq 0, \sum_{i \in A} \alpha_i = 1 \right\} \cap \left\{ \sum_{j \in B} \beta_j x_j \;\middle|\; \beta_j \geq 0, \sum_{j \in B} \beta_j = 1 \right\} \neq \phi$$

$$\Rightarrow \exists\, \alpha_i \geq 0, \sum_{i \in A} \alpha_i = 1 \;\&\; \beta_j \geq 0, \sum_{j \in B} \beta_j = 1$$

such that $$\sum_{i \in A} \alpha_i x_i = \sum_{j \in B} \beta_j x_j$$

$$\Rightarrow \sum_{i \in A} \alpha_i x_i - \sum_{j \in B} \beta_j x_j = 0$$

$$\Rightarrow \sum_{k \in A \cup B} \gamma_k x_k = 0$$

where $\gamma_k = \alpha_k \;\forall k \in A$ & $\gamma_k = -\beta_k \;\forall k \in B$

Lower bounding VC dim: If we find $d+1$ points $x_1, x_2, \ldots x_{d+1}$ that can be shattered by the class of halfplanes H, then

$$VCdim(H) \geq d+1.$$

Let the class H be defined by

$$H = \{ \text{sgn}(w^T x + b) \mid w \in \mathbb{R}^d, b \in \mathbb{R} \}$$

$$\text{where } \text{sgn}(z) = \begin{cases} +1 & \text{if } z \geq 0 \\ -1 & \text{if } z < 0 \end{cases}$$

Let us consider the $d+1$ points $\vec{0}, \vec{e}_1, \vec{e}_2 \cdots, \vec{e}_d$ which are the standard basis vectors with labels $y_0, y_1, y_2, \cdots y_d \in \{-1, +1\}$.

Now for each label vector $\vec{y} \in \{-1, +1\}^{d+1}$, we pick

$w_i = y_i$ & $b_i = \frac{1}{2} y_0$. Define $\underline{y} = [y_1 \cdots y_d]^T$

Then, $h(\vec{e}_i) = \text{sgn}(w^T e_i + b)$

$$= \text{sgn}\left(\underline{y}^T e_i + \frac{y_0}{2}\right) = \text{sgn}\left(y_i + \frac{1}{2} y_0\right)$$

$$= \text{sgn}(y_i) = y_i$$

Also $h(\vec{0}) = \text{sgn}(w^T 0 + b) = \text{sgn}(b)$

$$= \text{sgn}\left(\frac{1}{2} y_0\right) = \text{sgn}(y_0) = y_0$$

Hence all the configurations $y \in \{-1, +1\}^{d+1}$ can be realised by the halfplanes h in the class H.

Hence H shatters the $d+1$ points.
Hence $VC\dim(H) \geq d+1$

Now we want to show that $VC\dim(H) \leq d+1$
For that, we will show that no $d+2$ points can be shattered by H.

Let $x_1, x_2 \ldots x_{d+2}$ be arbitrary $d+2$ points in $\mathbb{R}^d$.
with binary labels $y_1, \ldots y_{d+2} \in \{-1, +1\}$

Assume that the above points are shattered by H

This means all labelings can be realized.

We apply Radon's theorem and find the index set partitions A & B such that $A \cap B = \phi$ & $A \cup B = \{1, 2, \ldots, d+2\}$

Let us look at a labeling where $y_i = +1$ if $i \in A$
& $y_i = -1$ if $i \in B$.

Let us assume this labelling can be realized by some $w_0 \in \mathbb{R}^d$, $b_0 \in \mathbb{R}$.

$\text{Sgn}(w_0^T x_i + b_0) = y_i \quad \forall i \in \{1, 2, \ldots, d+2\}$ by realization.

$\forall i \in A \quad \text{Sgn}(w_0^T x_i + b_0) = +1$

$\Rightarrow w_0^T x_i + b_0 \geq 0 \qquad \forall i \in A$

Similarly $w_0^T x_i + b_0 < 0 \quad \forall i \in B$

By Radon's theorem, there is non-empty intersection for the convex hull of $\{x_i \mid i \in A\}$ & $\{x_i \mid i \in B\}$

Let x be a point in

$$\text{ConvHull}\{x_i \mid i \in A\} \cap \text{ConvHull}\{x_i \mid i \in B\}$$

$\therefore \exists \alpha_i > 0, \sum_{i \in A} \alpha_i = 1$ such that $x = \sum_{i \in A} \alpha_i x_i$

Similarly $\exists\ \beta_i \geq 0$, $\sum_{i \in B} \beta_i = 1$ such that $x = \sum_{i \in B} \beta_i x_i$

Now, $w_0^T x + b_0 = w_0^T \sum_{i \in A} \alpha_i x_i + b_0$

$$= \sum_{i \in A} \alpha_i w_0^T x_i + b_0 \underbrace{\sum_{i \in A} \alpha_i}_{1}$$

$$= \sum_{i \in A} \alpha_i \underbrace{(w_0^T x_i + b_0)}_{+1}$$

$$= \sum_{i \in A} \alpha_i = +1$$

On the other hand,

$$w_0^T x + b = w_0 \sum_{i \in B} \beta_i x_i + b_0$$

$$= \sum_{i \in B} \beta_i w_0^T x_i + b_0 \underbrace{\sum_{i \in B} \beta_i}_{+1}$$

$$= \sum_{i \in B} \beta_i \underbrace{(w_0^T x_i + b_0)}_{-1}$$

$$= -\sum_{i \in B} \beta_i = -1$$

We now have a Contradiction !!!

Hence, the labeling Constructed using Radon's partitions are not realizable by $w_0 \in \mathbb{R}^d$, $b_0 \in \mathbb{R}$ & hence H cannot shatter any arbitrary $d+2$ points since the labeling constructed using Radon's partition on this set is not realizable by any halfspace $h \in H$.

(c) We will use the same technique as in part (b) to show that VC dim of ball classifiers in $\mathbb{R}^d$ is also $d+1$.

Balls in $\mathbb{R}^d$ is the set H

$$H = \{ Sgn(r - |\vec{x} - \vec{x}_0|_2) \mid x_0 \in \mathbb{R}^d, r \in \mathbb{R}, r > 0 \}$$

where $Sgn(z) = \begin{cases} +1 & \text{if } z \geq 0 \\ -1 & \text{if } z < 0 \end{cases}$

Lower Bound: $VC(H) \geq d+1$

We find a set of points $x_1, \ldots x_{d+1}$ that can be shattered by H.

Consider the same set of points as in part (b)

$$x_0 = \vec{0}, \; x_1 = \vec{e}_1, \; x_2 = \vec{e}_2, \ldots x_d = \vec{e}_d$$

Let $y_0, y_1 \ldots y_{d+1}$ be any labeling with $y_i \in \{-1, +1\}$

Lower Bound: $VC\dim(H) \geq d+1$

Define the spherical inversion map about any point x_0 as $\phi_{x_0}(x) = \begin{cases} \frac{x - x_0}{|x - x_0|^2}, & x \neq x_0 \\ 0, & x = x_0 \end{cases}$

Now, we observe that the inverse of this map is the map

$$\phi_{x_0}^{-1}(\vec{z}) = \begin{cases} \frac{z}{|z|^2} + x_0 & \text{if } z \neq 0 \\ x_0 & \text{if } \vec{z} = 0 \end{cases}$$

If $z = \vec{0}$, $\phi_{x_0} \circ \phi_{x_0}^{-1}(z) = \phi_{x_0}(x_0) = 0 = z$

If $x = x_0$, $\phi_{x_0}^{-1} \circ \phi_{x_0}(x_0) = \phi_{x_0}^{-1}(0) = x_0 = x$

If $x \neq x_0$,

$$\phi_{x_0}^{-1} \circ \phi_{x_0}(x) = \frac{\phi_{x_0}(x)}{|\phi_{x_0}(x)|^2} + x_0 = \frac{\frac{x-x_0}{|x-x_0|^2}}{\left|\frac{x-x_0}{|x-x_0|^2}\right|^2} + x_0$$

$$= \frac{x-x_0}{|x-x_0|^2} \cdot \frac{|x-x_0|^4}{|x-x_0|^2} + x_0 = x - x_0 + x_0 = x.$$

Again If $\vec{x} \neq 0$

$$\phi_{x_0} \circ \phi_{x_0}^{-1}(x) = \phi_{x_0}\left(\frac{x}{|x|^2} + x_0\right) = \frac{\frac{x}{|x|^2} + x_0 - x_0}{\left|\frac{x}{|x|^2} + x_0 - x_0\right|^2} = \frac{x}{|x|^2} \cdot \frac{|x|^4}{|x|^2} = x$$

From here on, we will only focus on the transform ϕ_0.

where $$\phi_0(x) = \begin{cases} \frac{x-0}{|x-0|} & x \neq 0 \\ 0 & x = 0 \end{cases} = \begin{cases} \frac{x}{|x|} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Now $$\phi^{-1}(z) = \begin{cases} \frac{z}{|z|} + 0, & z \neq 0 \\ 0, & z = 0 \end{cases} = \begin{cases} \frac{z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

Consider the ball classifiers $h_{(c,r)} \in H$ where

$$h_{(c,r)}(x) = \text{sgn}(r - |x-c|)$$

Consider the ball classifiers $h_{(c,|c|)}(x) = \text{sgn}(|c| - |x-c|)$

This ball classifier's decision boundary passes via origin.

If $Z = \phi_0(\mathbb{R}^d)$ be the image under ϕ_0, we want to find the boundary of the set of points labelled $+1$ & -1 under the classification by $h_{(c,|c|)}$ in the domain $\mathbb{R}^d$.

Boundary is the locus of $z \in Z$ such that $h_{(c,|c|)}(\phi_0^{-1}(z)) = 0$

The +ve labelled $z \in Z$ are $h_{(c,|c|)}(\phi_0^{-1}(z)) = +1$

$$\Rightarrow \text{Sgn}(|c| - |\phi_0^{-1}(z) - c|) = +1$$

$\Rightarrow \quad |c| - |\phi_0^{-1}(z) - c| \geq 0$

$\Rightarrow \quad |\phi_0^{-1}(z) - c|^2 \leq |c|^2$

$\Rightarrow \quad \left|\frac{z}{|z|^2} - c\right|^2 \leq |c|^2$ Assuming $z \neq \vec{0}$

$\Rightarrow \quad \frac{|z|^2}{|z|^4} + |c|^2 - 2\frac{c^T z}{|z|^2} \leq |c|^2$

$\Rightarrow \quad \frac{1}{|z|^2} - 2\frac{c^T z}{|z|^2} \leq 0$

$\Rightarrow \quad 1 - 2c^T z \leq 0 \quad \Rightarrow$

$\Rightarrow \quad c^T z - \frac{1}{2} \geq 0$

$\therefore$ In the image space $G_1 = \{ \operatorname{sgn}(c^T z - \frac{1}{2}) \mid c \in \mathbb{R}^d \}$ are the halfspace classifiers corresponding to the ball classifiers passing via origin in the original space.

Now, if we show that the set of classifiers $Q \subset H$ where $Q = \{ h_{(c, |c|)} \mid c \in \mathbb{R}^d \}$ are ball classifiers restricted to the ones passing via origin can shatter $d+1$ points, we have $VC(Q) \geq d+1$

& since $H \supset Q$, $VC(H) \geq VC(Q) \geq d+1$

Using part (b) we claim that there exist $d+1$ points in the image space Z that can be shattered by halfspaces (assume non of these halfspaces pass via origin).

Now for each halfspace of the form $w^T z + b = 0$, multiply by $-\frac{1}{2b}$ to get $\left(\frac{w}{2b}\right)^T z - \frac{1}{2} = 0$

Hence these hyperplanes are sphere in original space & can shatter the preimages of the $d+1$ points.

* Note: I still think the above arguement isn't rigorous enough !!!

Upper Bound : $VC(H) \leq d+1$

We will show that no $d+2$ points can be shattered by H.

For the $d+2$ points, do the labeling as done in part (b) using Radon's partitions.

Now, let the index set A & B be the Radon partition such that $A \cup B = \{1, \ldots d+2\}$ & $A \cap B = \phi$.

$$\forall i \in A, \; y_i = +1 \quad \& \quad \forall i \in B, \; y_i = -1$$

Now, let there exist a ball such that it classifies all x_i as $+1$ if $i \in A$ & x_i as -1 if $i \in B$

$$h_{(c,r)}(x_i) = \text{Sgn}(r - |x_i - c|) = +1 \quad \forall i \in A$$

$$\Rightarrow \quad |x_i - c| < r \quad \forall i \in A$$

Since this ball is convex & contains all the points $\{x_i \mid i \in A\}$ within r distance away from its center, the convex hull of this set is also in the Ball & hence must be classified $+1$.

By Radon's theorem,

let $x \in \text{ConvHull}(\{x_i \mid i \in A\}) \cap \text{ConvHull}(\{x_i \mid i \in B\})$

$$h(x) = \text{Sgn}\left(\gamma - \left| \sum_{i \in A} \alpha_i x_i - c \right| \right)$$

$$= \text{Sgn}\left(\gamma - \left| \sum_{i \in A} \alpha_i (x_i - c) \right| \right)$$

$$\gamma - \left| \sum_{i \in A} \alpha_i (x_i - c) \right| \geq \gamma - \sum_{i \in A} \alpha_i |x_i - c|$$

But $|x_i - c| \leq \gamma \quad \forall i \in A$

$$\Rightarrow \alpha_i |x_i - c| \leq \alpha_i \gamma \quad \forall i \in A$$

$$\Rightarrow \sum_{i \in A} \alpha_i |x_i - c| \leq \sum_{i \in A} \alpha_i \gamma = \gamma \underbrace{\sum_{i \in A} \alpha_i}_{1} = \gamma$$

$$\Rightarrow \sum_{i \in A} \alpha_i |x_i - c| \leq \gamma$$

$$\Rightarrow -\sum_{i \in A} \alpha_i |x_i - c| \geq -\gamma$$

$$\Rightarrow \gamma - \sum_{i \in A} \alpha_i |x_i - c| \geq 0$$

$$\Rightarrow \text{Sgn}\left(\gamma - \sum_{i \in A} \alpha_i |x_i - c| \right) = +1$$

$$\Rightarrow h(x) = +1$$

Now, $x \in \text{Conv Hull}\{ x_i \mid i \in B \}$

$$\therefore x = \sum_{i \in B} \beta_i x_i \quad , \quad \beta_i \geq 0, \quad \sum_{i \in B} \beta_i = 1$$

$$h(x) = \text{Sgn}\left(\gamma - |x - c| \right)$$

$$= \text{Sgn}\left(\gamma - \left| \sum_{i \in B} \beta_i x_i - c \right| \right)$$

$$\left| \sum_{i \in B} \beta_i x_i - c \right| = \left| \sum_{i \in B} \beta_i (x_i - c) \right|$$

$$\leq \sum_{i \in B}' \beta_i (x_i - c)$$

— — — —

Proceed later.